

## The way to LEAD (Lasting and Effective AI Dominance)

To whom it may concern:

The United States is poised to become the undisputed global leader in AI, but only if it pivots away from the past AI strategy. Like the industrial revolution, the AI revolution can redraw the balance of power in the international order. Europe's regulatory environment is stifling their ability to compete while China's authoritarian system is accelerating their path to strong intelligence. Recent Chinese releases of competitive large language models underscore the urgent timeline required to outpace our geopolitical adversaries in achieving **lasting** global dominance in AI.

We believe actions underpinning U.S. global dominance in AI manifest as:

1. **Global Adoption:** American technology leads AI adoption at home and abroad  
**To Do:** Build trust in American-led technology by leading in AI standards, and tools to validate AI products against these standards.
2. **Strongest Intelligence:** American AI outperforms competitors in all industries.  
**To Do:** Maximize the availability of high quality data to AI systems.  
Mitigate hallucination through data citation and data provenance.
3. **Abundant Data:** the American data economy is the world's leading marketplace.  
**To Do:** Create incentives for investment in and sharing of data and information.
4. **National Security:** U.S. AI policy mitigates risks and maximises innovation edge  
**To Do:** Achieve resilience to an AI-facilitated attack by limiting system centralization and implementing other elements of the AI fire code.
5. **Command of Talent:** The U.S. dominates access to compute power and talent  
**To Do:** Investment in AI innovation centers will draw AI talent to our shores.

Our initiatives:

- **AI Innovation Centers:** The Æthos Foundation has created the largest AI Start up hub in Boston and is beginning to expand its model around the U.S.
- **Smart Regulation:** Æthos tracks products' alignment with data and AI best practices to help developers build trust to access proprietary sensitive data.
- **Data Loops:** Data.Flowers creates intelligence amplification technology for tracking media and public health.

Who we are:

- **Sam Rowe:** Æthos Founder & CEO, Sam recently consulted for the Massachusetts governor on AI innovation policy and studied Science Technology and International Affairs at Georgetown School of Foreign Service.
- **Aleks Jakulin:** Data.Flowers Founder & CEO, Æthos advisor, Aleks received the EurAI award for the best PhD dissertation in artificial intelligence in Europe, and coauthored standards like PNG and JPEG used by billions.

## A) Investment in Data is needed for the Next Wave of AI

Ilya Sutskever, the cofounder and former chief scientist of OpenAI, stated that what's been the driver for all the progress we see today, including OpenAI's GPT-3 -- have been extraordinarily large neural networks trained on a huge dataset: the internet. While the computing power has been growing, data itself has not been growing. He added that data is the fossil fuel of AI.

<https://www.appblit.com/scribe?v=1yvBqasHLZs&t=475&g=5DOebERzwDMcSHkSGIRsfqdfunG2>

### More data will lead to more intelligence!



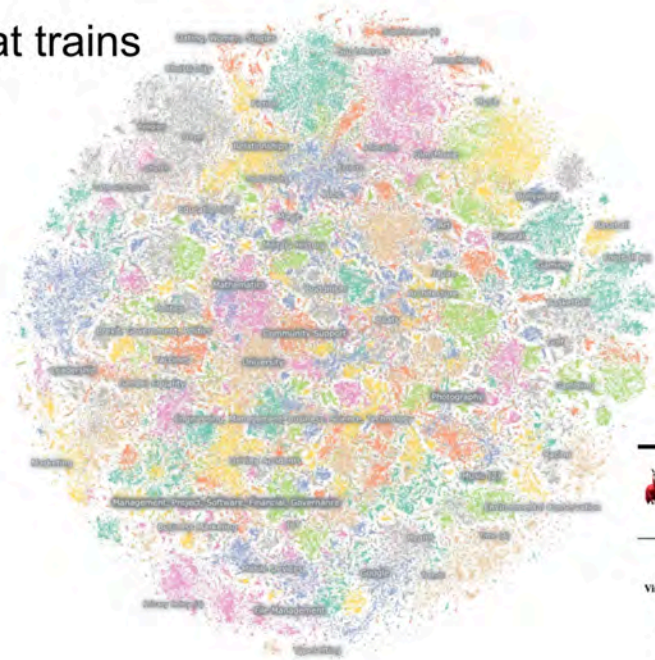
But we need:

1. Data needs to be **explicit** (verifiability, truth)
2. Data needs to be **licensed** (media, reference, science)
3. Data needs to be **excludable** (privacy, safety, censorship)

With our own data controls...

Right now, the incentive to invest in data is limited. Most of the value of data is captured by the services that harvest it. The price for data is falling. Buyers of data compete with data collectors and curators. There has been virtually no new successful entrants into the data ecosystem. There have been several examples of past failure to protect the data ecosystems:

## Data that trains the AI



### RedPajama: an Open Dataset for Training Large Language Models

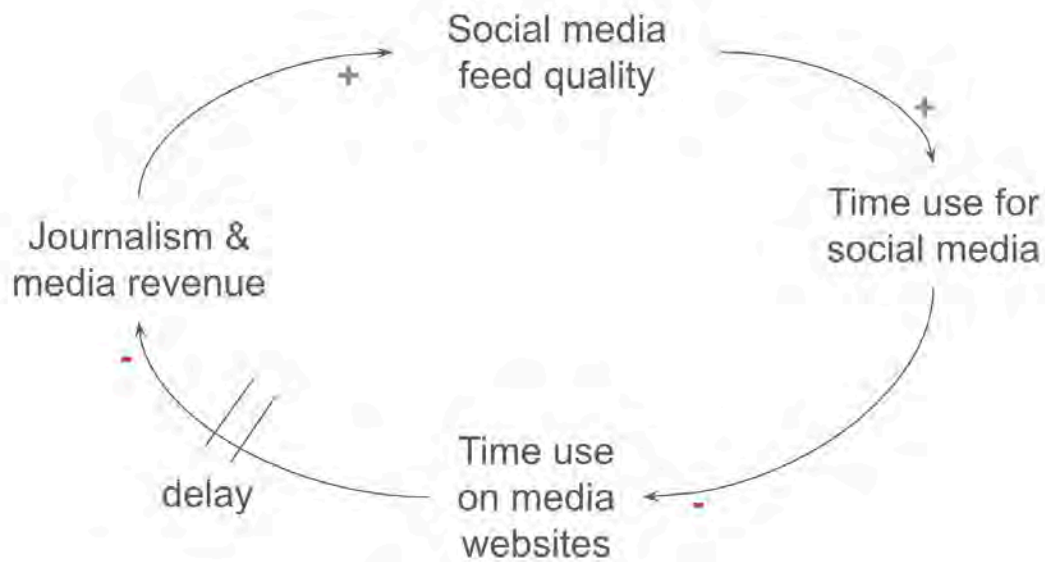
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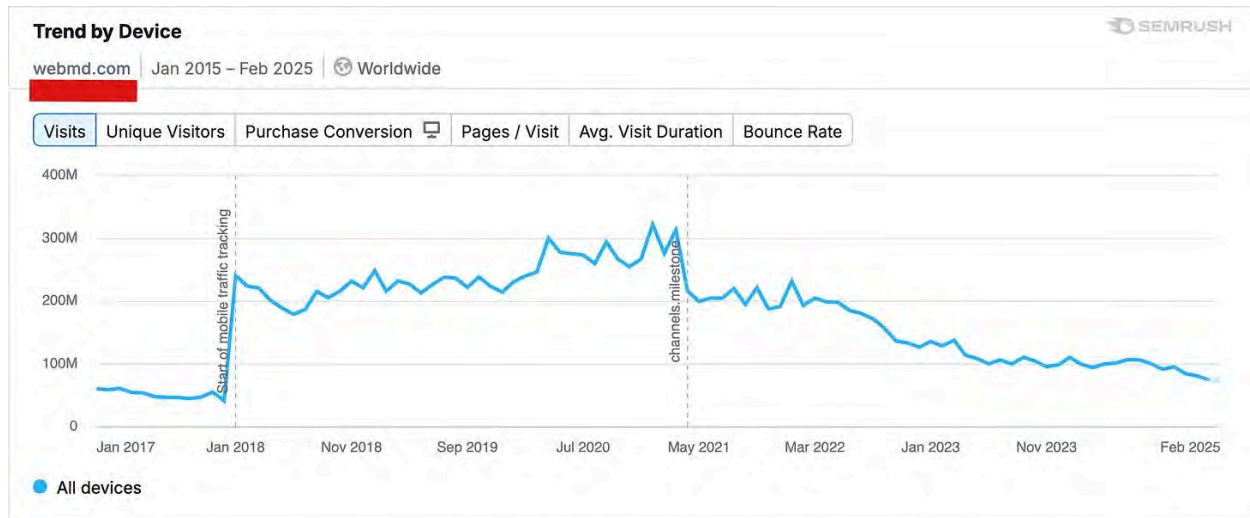
1. Yahoo curated and categorized websites. Google's search leveraged the curation by Yahoo. Once the market for search switched to Google, Yahoo's service was less desirable, and the data was no longer updated. As a result, Google search results have been deteriorating. Google has been compensating by utilizing the search behavior of its users to improve its own search results, but this data was not usable by researchers or other competitors, leading to overall deterioration not just of search but of the internet as a whole.



2. Media companies have been creating content that then fueled the rise of social media. Social media platforms initially routed traffic back to media companies, but with time this relationship has been eroding, with decades of falling media sector revenue. The quality of media has deteriorated, and new "influencer" media operations have resulted in more political polarization and less accurate information in general. With the adoption of AI-powered tools, this negative trend is likely going to accelerate.



3. The AI systems used for software development have been trained on communities like Stack Overflow. With the rise of AI-powered tools, the participation in Stack Overflow has been declining, leading to a decrease in the quality of the data used to train AI systems. Similar patterns of deterioration have been identified for other valuable publishers and communities, such as WebMD for medical information. As with Google in the past, the interaction data is held within individual programming tools, leading to a winner-take-all model.



A further challenge is hallucination, which is the ability of AI systems to generate content that is not based on any real-world data. There is a clear solution: AI systems should generate output that cites underlying data. Data citation solves two problems: it provides an accounting basis for the data licensing business model. Furthermore, it addresses the AI hallucination problem, by ensuring that AI systems are transparent about their sources of information - and giving a foundation for AI explainability.

**Policy recommendation:** Establish a data commons office that ensures that investing in data yields solid returns. Ensure that the contributions to the world data economy are balanced instead of some countries free-riding on the benefits of data commons.

**Outcome:** This will lead to the second wave of innovation in artificial intelligence resulting from further public, private, entrepreneurial, and international investment into data commons.

## B) The AI Risks are Best Handled with Fire Codes

AI technology can be considered as an offensive weapon. Our information technology is not yet fully robust against AI-powered attacks. It's a matter of national security to proof our nation against them.

There have been two views of AI technology. One is that AI is a complex centralized technology like nuclear, where it's possible to control dissemination. The other is that AI is a decentralized technology which will rapidly spread around the world, like fire - and where rather than attempting to control the technology one should control the resilience towards technology. For example, we control fire through fire codes and fire departments, because it's too easy to create fire.

AI has been seen as a centralizing technology – where the data and the knowledge are assembled, to use Sam Altman, OpenAI founder's words, in a "skyscraper full of computers". Such a centralized system is highly vulnerable to a takeover in an adversarial environment, and the narrative over the past decade has been that such systems should be closely affiliated with governments, in many ways extensions of political leadership with strong ideological affinities.

With the recent developments around the world including, for example, DeepSeek, it's becoming clear that AI is not a centralizing technology. It's instead better viewed as something like fire: there are people all around the world who can use large language models, for either good or bad. It can't be put back into the bottle and controlled. Instead, we should think about how the world adapted to other similar innovations, like fire, metal, or plastics.

Perhaps the best way to view AI risks is through fire codes. While explosive devices and highly combustible materials can to some extent be controlled, like high performance GPUs – enough damage can be created by uncontrollable off-the-shelf computing devices. This is not new: over the past couple of decades, we've seen increasing levels of spam, cybersecurity risks, impersonation, and social engineering attacks.

We regulate infrastructure not to be destroyed by fire:



There are two fundamental threats exacerbated by AI technology:

- **Take-over:** a compromised individual or small team amplified by AI would be able to move rapidly through systems, and centralized systems with no or few humans in the loop could be taken over almost immediately. For example, a large

data center that's internally unencrypted or encrypted with a small number of master keys could be monitored by an external party without detection - and when the time is right, taken over at once. An operating system could be remotely updated, breaking a considerable percentage of all computers running that operating system.

- **Impersonation:** AI technology advances from the last few years are now breaking authentication and proofs-of-humanity. A compromised operating system that deploys agents to personal and company computers could in turn compromise anything that requires mere keystrokes or digital signatures.

Individuals and companies do care about cybersecurity, but not at the level of being resilient to this type of an attack. It's the government's job to ensure ecosystem-level resilience by regulating the minimum security of the devices in the market. To some extent, this has been done by limiting the market share of certain manufacturers, but we recommend a more structured approach.

Even at a smaller level, a free market requires companies and individuals to retain some privacy. In the absence of privacy and competitive advantage by an innovator is quickly lost to the economies of scale of large competitors. As a result, if we want to retain the value of entrepreneurial dynamism, it is important to protect the privacy of small enterprises from larger platform operators.

Such an approach would have five prongs:

**Decentralization:** the same way that the US Constitution separated the governance between multiple branches of the government, we need to ensure that our technological dependencies aren't excessively centralized. Vertical integrations between hardware, operating systems, applications software, data, and commerce are conflicting with an efficient market economy open to entrepreneurship. We need decoupling and we need to assume that any organization can be taken over by a hostile adversary.

**Modularization:** the complexity and entanglement of software is unacceptably modularized. The same computer runs both banking software as well as downloaded computer games. The same software environment powers both encryption as well as software development. Most software has levels of complexity that make it almost impossible to verify the absence of security holes. We need much simpler devices and clearer isolation principles that allow rapid innovation of software but isolate high security operations. The ability of any device to spy on the customer, to update its software, or to facilitate other forms of external control is unacceptable, and we need



standards that introduce friction to such situations, especially for devices that are no longer rapidly evolving.

**Hard Decentralized Authentication:** Users are going to be activating and authorizing AI agents to act on their behalf. This will be both a boon to productivity, but also a massive opening for attacks. The solution lies in clearly separating what can be delegated to a computer – and that which requires an in-person signoff. Hard authentication through a new type of device is how the rate of spread of an attack is limited.

**Game-Theoretic Mechanism Design:** Instead of relying on centralized systems to detect spam or AI-generated disinformation, we should instead seek to develop market incentives that ensure that bad actors can't be better off than good actors. Instead of relying on centralized services that detect spam, we should instead encourage the development of microescrow systems that require senders of email to hold funds in reserve in case the customers decide that messages are spam. Instead of trying to detect if information is AI-generated, we should encourage information provenance where reporters and photographers are cited by the content, along with the whole chain of modifications to the original time-stamped information.

**Market Efficiency instead of Market Control:** It's not necessary to exert market control to control the market. Instead of trying to regulate the products and prices, a marketplace can simply provide accurate scales that lets buyers and sellers measure the weight of the sold products without disagreements. Instead of trying to control information sources, a government can simply provide data about the information sources, letting the market participants act on it. Instead of trying to censor disinformation bots, a government can provide data about information sources. Instead of trying to police information, a government could simply allow for information retraction notices to be propagated.

**Policy recommendations:**

1. Create a red teaming group systematically identifying attacks on companies, individuals and state actors. When a potential exploit is found, the underlying system needs to be broken down and decoupled, fixing the exploit is not enough.
2. Drive for development of modular architectures that will contain an exploit within a particular module without contagion.
3. Develop next generation human-speed authentication systems for the US military and government needs.



4. Evaluate incentive systems and propose adjustments when they're vulnerable to AI-enabled adversarial actors.

5. Establish data loops that evaluate the exposure of the US citizens to adversarial influence information operations with a budget that's proportional to adversaries' investment in information operations (~\$1-10B annual).

**Outcome:** U.S. becomes resilient to AI-powered attacks within 2 years.

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